M3503 WEEK 5 DEC 2021

# REDUCTION OF ORDER FOR SECOND ORDER HOMOGENEOUS EQUATIONS

Consider the second order homogeneous equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \quad \text{—①}$$

Let $y_1$ be a nontrivial solution to ① on an interval I. The second linearly independent solution of ① is $y_2 = y_1 v(x)$, where $v(x)$ can be determined by substituting $y_2$ into equation ①. This substitution reduces the equation to a first order variable separable equation in $v'$. The general solution is $y = c_1 y_1 + c_2 y_2$

Examples

$y_1$ is the solution to the given equation, use reduction of order to find the general solution

① $y'' - 4y' + 4y = 0$ ; $y_1 = e^{2x}$

② $y'' - y = 0$ ; $y_1 = \cosh(x)$

Sol^n

① Let $y_2 = ve^{2x}$

$$\Rightarrow y_2' = 2ve^{2x} + v'e^{2x}$$

$$\Rightarrow y_2'' = 4ve^{2x} + 2v'e^{2x} + 2v'e^{2x} + v''e^{2x}$$
$$= v''e^{2x} + 4v'e^{2x} + 4ve^{2x}$$

Substituting into the equation, we get

$v''e^{2x} + 4v'e^{2x} + 4ve^{2x} - 4v'e^{2x} - 8ve^{2x} + 4ve^{2x} = 0$

$\Rightarrow v''e^{2x} = 0$

$\Rightarrow u'e^{2x} = 0$ , where $u = v'$

$\Rightarrow u' = 0$

$\Rightarrow u = k_1 = v'$

$\Rightarrow v = k_1 x + k_2$

$\therefore y = c_1 e^{2x} + c_2 k_1 x e^{2x} + c_2 k_2 e^{2x}$

$= Ae^{2x} + Bxe^{2x}$

(2): let $y_2 = v\cosh(x)$

$\Rightarrow y_2' = v'\cosh(x) + v\sinh(x)$

$\Rightarrow y_2'' = v''\cosh(x) + 2v'\sinh(x) + v\cosh(x)$

Substituting, we get

$v''\cosh(x) + 2v'\sinh(x) + v\cosh(x) - v\cosh(x) = 0$

$\Rightarrow v''\cosh(x) + 2v'\sinh(x) = 0$

$\Rightarrow u'\cosh(x) + 2u\sinh(x) = 0$ , where $u = v'$

$\Rightarrow u' + 2u\tanh(x) = 0$

$\Rightarrow \frac{u'}{u} = -2\tanh(x)$

$\Rightarrow \ln u = -2\ln(\cosh(x)) - \ln c$

$\Rightarrow \ln(cu) = \ln[(\cosh(x))^{-2}]$

$\Rightarrow cu = \operatorname{sech}^2(x)$

$\Rightarrow u = c_1 \operatorname{sech}^2(x)$

$\Rightarrow v' = c_1 \operatorname{sech}^2(x)$

$\Rightarrow v = c_1 \tanh(x) + c_2$

$\therefore y = c_1\cosh(x) + c_3 c_1 \tanh(x)\cosh(x) + c_2 c_3 \cosh(x)$

$= A\cosh(x) + B\sinh(x)$

# Reduction Of Order For Second Order Non-Homogeneous Equations

Consider the second order equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = f(x) \quad , f(x) \neq 0 \quad \text{---} ①$$

Suppose $y_1$, defined on interval $I$, is a nontrivial solution to the homogeneous equation associated with equation ①. Substituting $y_2 = y_1 v$ into ① reduces the equation to a first order linear equation in $v'$. The general solution to equation ① is $y = y_2$.

Examples

$y_1$ is the solution to the associated homogeneous equation. Find the general solution of the given eqⁿ

① $y'' - 4y = 2$ ; $y_1 = e^{-2x}$

② $y'' - 3y' + 2y = 5e^{3x}$ ; $y_1 = e^{x}$

solⁿ

① let $y_2 = v e^{-2x}$

$\Rightarrow y_2' = v'e^{-2x} - 2ve^{-2x}$

$\Rightarrow y_2'' = v''e^{-2x} - 4v'e^{-2x} + 4ve^{-2x}$

Substituting into the equation, we get

$v''e^{-2x} - 4v'e^{-2x} + 4ve^{-2x} - 4ve^{-2x} = 2$

$\Rightarrow v''e^{-2x} - 4v'e^{-2x} = 2$

$\Rightarrow u'e^{-2x} - 4ue^{-2x} = 2$ , where $u = v'$

$\Rightarrow u' - 4u = 2e^{2x}$

Integrating factor $= \exp\int -4\,dx$
$= e^{-4x}$

So, $e^{-4x}u' - 4ue^{-4x} = 2e^{-2x}$
$\Rightarrow \frac{d}{dx}(ue^{-4x}) = 2e^{-2x}$
$\Rightarrow ue^{-4x} = -e^{-2x} + C_1$
$\Rightarrow v' = -e^{2x} + Ce^{4x}$
$\Rightarrow v = -\frac{1}{2}e^{2x} + C\frac{1}{4}e^{4x} + C_2$

$\therefore y = -\frac{1}{2} + C\frac{1}{4}e^{2x} + C_2e^{-2x}$
$= Ae^{2x} + Be^{-2x} - \frac{1}{2}$

(2) let $y_2 = ve^x$
$\Rightarrow y_2' = v'e^x + ve^x$
$\Rightarrow y_2'' = v''e^x + 2v'e^x + ve^x$

Substitution into the equation gives

$v''e^x + 2v'e^x + ve^x - 3v'e^x - 3ve^x + 2ve^x = 5e^{3x}$
$\Rightarrow v''e^x - v'e^x = 5e^{3x}$
$\Rightarrow u' - u = 5e^{2x}, \quad u = v'$

Integrating factor $= \exp(\int -1\,dx)$
$= e^{-x}$
So, $\frac{d}{dx}(ue^{-x}) = 5e^x$
$\Rightarrow ue^{-x} = 5e^x + C_1$
$\Rightarrow v' = 5e^{2x} + Ce^x$
$\Rightarrow v = 5/2\,e^{2x} + Ce^x + C_2$
$\therefore y = 5/2\,e^{3x} + Ae^{2x} + Be^x$

## Reduction Of Order In General

Reduction of order can be used to solve any higher order differential equation, homogeneous or non-homogeneous.

For $n^{th}$ order equation, if $y_1$, which is the solution of the associated homogeneous equation, is known, the equation can be reduced to an $(n-1)^{th}$ order equation in $v^{(n-1)}$.

If the solution to the homogeneous equation associated with the resulting equation is known, the process can be repeated until the solution is found.

# Inverse Operator Methods

Consider a linear differential equation with constant coefficients

$a_n D^n y + a_{n-1} D^{n-1} y + \cdots + a_1 D y + a_0 y = f(x)$, $f(x) \neq 0$.

Which can be written as

$$F(D)\, y = f(x) \quad \text{—①}$$

where $F(D) = a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0$ is a differential operator.

From, we get the particular solution as

$$y_p = \frac{1}{F(D)} f(x),$$

where $\frac{1}{F(D)}$ is an inverse differential operator.

Using the following useful formulae, we apply the inverse method to solve equations with specific forms of non-homogeneous term $f(x)$.

① $F(D)[e^{ax} V(x)] = e^{ax} F(D+a)\, V(x)$

This is known as shift formula

② $\frac{1}{F(D)} e^{ax} = \frac{e^{ax}}{F(a)}$, if $F(a) \neq 0$

If $F(a) = 0$, then $F(D) = (D-a)^h G(D)$, $G(a) \neq 0$

So

$$\frac{1}{F(D)} e^{ax} = \frac{1}{(D-a)^h G(D)} e^{ax}$$

$$= \frac{1}{G(a)} \left[ \frac{1}{(D-a)^h} e^{ax} \cdot 1 \right]$$

$$= \frac{e^{ax}}{g(a)} \frac{1}{D^n}(1) \quad \text{by shift formula}$$

$$= \frac{e^{ax}}{g(a)} \frac{x^n}{n!}$$

3. Recall Euler's formula

$$e^{i(ax+b)} = \cos(ax+b) + i\sin(ax+b)$$

$$\frac{1}{F(D)} \cos(ax+b) = Re\left[\frac{1}{F(D)} e^{i(ax+b)}\right] \quad \text{and}$$

$$\frac{1}{F(D)} \sin(ax+b) = Im\left[\frac{1}{F(D)} e^{i(ax+b)}\right]$$

Where "Re" is real part of, and "Im" is imaginary part of.

4. $\frac{1}{F(D)} P(x) = \frac{1}{1+g(D)} P(x)$, where $P(x)$ is a polynomial

- Then expand $\frac{1}{1+g(D)}$ by binomial expansion

Binomial series; for any real number $K$

$$(1+x)^K = \sum_{n=1}^{\infty} \binom{K}{n} x^n = 1 + Kx + \frac{K(K-1)}{2!}x^2 + \frac{K(K-1)(K-2)}{3!}x^3 + \cdots$$

Alternatively to binomial expansion, use long division to work out $1+g(D) \overline{)P(x)}$

N.B Unlike normal long division, start division with 1 (a constant term), instead of the term with the highest degree of $D$

Examples

Find a particular solution to

① $(D^2-4D+3)y = e^{2x}$

② $(D^3+1)y = 3+e^{-x}+5e^{2x}$

③ $(D^4+10D^2+9)y = \cos(2x+3)$

④ $(D^4-1)y = \sin x$

⑤ $(D^2-4)y = \cosh(2x-1)+3^x$

⑥ $(D^3+D^2+1)y = x^2+2x+1$

⑦ $(D^3+D^2+D+1)y = x^4+2x+1$

Soln

① $(D^2-4D+3)y = e^{2x}$

$\Rightarrow y_p = \frac{1}{D^2-4D+3} e^{2x}$

$= \frac{1}{4-8+3} e^{2x}$

$= -e^{2x}$

② $(D^3+1)y = 3+e^{-x}+5e^{2x}$

$\Rightarrow y_p = \frac{1}{D^3+1}[3+e^{-x}+5e^{2x}]$

$= \frac{1}{D^3+1} 3e^{0x} + \frac{1}{(D+1)(D^2-D+1)} e^{-x} + \frac{1}{D^3+1} 5e^{2x}$

$= 3 + \frac{x e^{-x}}{3} + \frac{5e^{2x}}{9}$

③ $(D^4 + 10D^2 + 9)\, y = \cos(2x+3)$

$$\Rightarrow y_p = \frac{1}{D^4+10D^2+9} \cos(2x+3)$$

$$= \mathrm{Re}\left[\frac{1}{D^4+10D^2+9} e^{i(2x+3)}\right]$$

$$= \mathrm{Re}\left[e^{i3} \frac{1}{D^4+10D^2+9} e^{i2x}\right]$$

$$= \mathrm{Re}\left[\frac{e^{i3}\, e^{i2x}}{16-40+9}\right]$$

$$= \mathrm{Re}\left[\frac{-e^{i(2x+3)}}{15}\right]$$

$$= \mathrm{Re}\left[\frac{-1}{15}\left[\cos(2x+3) + i\sin(2x+3)\right]\right]$$

$$= -\frac{1}{15}\cos(2x+3)$$

④ $(D^4-1)\, y = \sin x$

$$\Rightarrow y_p = \frac{1}{D^4-1} \sin x$$

$$= \mathrm{Im}\left[\frac{1}{D^4-1} e^{ix}\right]$$

$$= \mathrm{Im}\left[\frac{1}{(D^2-1)(D^2+1)} e^{ix}\right]$$

$$= \mathrm{Im}\left[\frac{-1}{2} \frac{1}{(D-i)(D+i)} e^{ix}\right]$$

$$= \mathrm{Im}\left[\frac{-1}{4i}\left(\frac{1}{D-i} - \frac{1}{D+i}\right) e^{ix}\right] \text{ by partial fractions}$$

$$= \mathrm{Im}\left[\frac{-1}{4i}\left(x e^{ix} - \frac{e^{ix}}{2i}\right)\right] = \mathrm{Im}\left[\frac{ixe^{ix}}{4} - \frac{1}{8} e^{ix}\right]$$

$$= \frac{x}{4}\cos x - \frac{1}{8}\sin x$$

⑤ $(D^2-4)y = \cosh(2x-1) + 3^x$

$\Rightarrow y_p = \frac{1}{D^2-4}\left[\cosh(2x-1) + 3^x\right]$

$= \frac{1}{D^2-4}\left[\frac{e^{2x-1}+e^{-(2x-1)}}{2} + e^{x\ln 3}\right]$

$= \frac{1}{2}\left[e^{-1}\frac{1}{(D-2)(D+2)}e^{2x} + e\frac{1}{(D-2)(D+2)}e^{-2x}\right] + \frac{1}{D^2-4}e^{x\ln 3}$

$= \frac{1}{2}\left[\frac{e^{-1}}{4}e^{2x}x + \frac{e}{-4}xe^{-2x}\right] + \frac{1}{(\ln 3)^2-4}e^{x\ln 3}$

$= \frac{1}{4}\left(\frac{e^{2x-1} - e^{-(2x-1)}}{2}\right) + \frac{3^x}{(\ln 3)^2-4}$

$= \frac{1}{4}\sinh(2x-1) + \frac{3^x}{(\ln 3)^2-4}$

⑥ $(D^3+D^2+1)y = x^2+2x+1$

$\Rightarrow y_p = \frac{1}{D^3+D^2+1}x^2+2x+1$

$= (1-(D^3+D^2)+\dots)(x^2+2x+1)$ binomial expansion

$= x^2+2x+1 - D^3(x^2+2x+1) - D^2(x^2+2x+1) + 0$

$= x^2+2x+1 - 0 - 2$

$= x^2+2x-1$

OR

$$
\begin{array}{r|l}
 & x^2+2x-1 \\
\hline
1+D^2+D^3 & x^2+2x+1 \\
 & -(x^2+2+0) \\
\hline
 & 2x-1 \\
 & -(2x+0) \\
\hline
 & -1 \\
 & -(-1) \\
\hline
 & 0
\end{array}
$$

$\therefore y_p = x^2+2x-1$

(7) $(D^3+D^2+D+1)y = x^4+2x+1$

$\Rightarrow y_p = \frac{1}{D^3+D^2+D+1}\, x^4+2x+1$

$$
\begin{array}{r|l}
 & x^4-4x^3+2x+23 \\
\hline
1+D+D^2+D^3 & x^4+2x+1 \\
 & -(x^4+4x^3+12x^2+24x) \\
\hline
 & -4x^3-12x^2-22x+1 \\
 & -(-4x^3-12x^2-24x-24) \\
\hline
 & 2x+25 \\
 & -(2x+2) \\
\hline
 & 23 \\
 & -(23) \\
\hline
 & 0
\end{array}
$$

$\therefore y_p = x^4-4x^3+2x+23$